ONE PAGE · PREPARED FOR TRIPADVISOR DATA PLATFORM · SENIOR DATA ENGINEER

Every event counted once, **on time, and checked.**

Kafka in, Flink in the middle, BigQuery out: NEXUS, the platform I built at Lifepal, applied to a travel site's events.

Kafka topics → an event-time stream job (Python, with a Java twin) → a nightly Spark batch that checks it → one contract rendered to Snowflake and BigQuery → SLAs with an incident and its postmortem. Synthetic data; everything runs from **rfim.github.io/nexus-tripadvisor**.

BUILT IN lifepal → PRESENTED TO Tripadvisor

PROVEN AT LIFEPAL

from my CV

24h → <5m reporting latency **6h → 90m** nightly batch **$11k** vs $19k forecast **500+** quality checks **−60%** data-trust escalations **3–5** engineers led, 2 promoted

THE LAST RUN

a week, 5–11 October

26,426
Kafka records, 13 partitions, three topics

0
stream/batch differences unexplained, of 9,748 windows

11/11
warehouse checks pass, masking and erasure included

79 min
the one SLA breach; 74 restarts; postmortem written

SIX PROBLEMS, PLANTED AND CAUGHT

each tested

PROBLEM	IN THE RUN	WHAT HANDLES IT
SDK retries	112 events re-sent, same event_id, new offset	Dedupe on id, 24 h checkpointed state: 112 dropped
An offline upload	192 iOS events, 2–8 hours late	Side output past an hour's lateness; the batch explains 159 windows
A schema change	app 7.4: price {amount, currency}, 302 records	One reader upcasts v1 and v2; unknown currency is a dead letter
A poison pill	partition 3, offset 171: 57 bytes of truncated JSON	74 restarts, paged at 14:35; fixed with the DLQ
A scraper	1,515 views in an hour that has 29.0	Hour held (2σ and ×2); 3 users removed; released
Erasure, then a stale version	an outbox replay re-sends version 3 after the tombstone	Version rule: 1 rejected; deleted in both warehouses

THE ROLE, AND WHERE IT SHOWS

stated plainly

ASK	HERE
Snowflake · BigQuery	warehouse/ from one contract
SLAs · RCA	sla.py, the incident page
Kafka · Spark	stream.py, spark_batch.py
Java	WindowJob.java, identical to Python
Flink	flink/job: real Flink 1.20 in CI, identical windows; not in production, said so

QUESTIONS I'D BRING

for Data Platform

1. Which pipelines page most, and why?
2. How are freshness SLAs measured today?
3. Where do Snowflake and BigQuery each fit?
4. How is streaming split between Flink and Spark?
5. What would a great first 90 days look like?

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